
The Factor of Two

Light bending as the fingerprint of entropic gravity — an essay on gravitation (~1,450 words)

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Provenance disclosure: this document was constructed by four AI models (Claude Fable 5 — synthesis and implementation; DeepSeek-V4-Pro — derivations; GLM-5.2 — refereeing; Kimi-K2.7 — numerics) in an adversarial-review protocol, with a human originator (B. Kloosterman) supplying the thesis and direction. Every claim's attack history, including retractions by all four models, is preserved in the accompanying repository. Numerical results are reproducible from committed scripts. Treat accordingly.

The Factor of Two: Light Bending as the Fingerprint of Entropic Gravity

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In 1911, Einstein computed the deflection of starlight by the Sun and got 0.87 arcseconds. The calculation was honest and the physics was beautiful: clocks run slower in a gravitational potential, wavefronts tilt where their crests age at different rates, and light bends toward the mass. It was also wrong by exactly a factor of two. The 1915 theory doubled it, Eddington measured 1.75" in 1919, and the difference between those numbers has been quietly burying theories ever since. Any account in which gravity is *only* differential time flow — Nordström's scalar theory was the cleanest casualty — predicts the 1911 number. The 1919 number requires curved space in equal measure with curved time, and it requires the two curvatures locked at equal strength.

We want to persuade you that this old factor of two is the observational fingerprint of a specific idea: that gravity and time are two readings of a single ledger, and the ledger is entropy.

The idea is no longer purely philosophical at either end. On the time side, a 2026 cold-atom experiment (Barontini, Phys. Rev. Research 8, L022047) realized a “mini-universe”: an isolated condensate with a time-independent Hamiltonian, partitioned by a barrier into an observed and an unobserved sector. With no external clock allowed, an internal time defined by coarse-grained entropy exchange between the sectors robustly orders the observed sector's expansion–recollapse cycles; raise the barrier and time itself slows, then stops. Time — its flow, its rate, its arrow — was read entirely as entropy bookkeeping, in a laboratory. On the gravity side, an established lineage (Jacobson 1995; Faulkner et al. 2014; Jacobson 2016) derives the Einstein equations from the first law of entanglement, $\delta S = \delta \langle K \rangle$, applied to local causal horizons, and a recent result sharpens what exactly that input buys: thermodynamic derivations of gravity yield precisely the *trace-free* Einstein equations — unimodular gravity — with the cosmological constant entering only as an integration constant (Alonso-Serrano & Liška 2022).

Here is the observation we build on. Entropy, temperature, and geometry meet operationally in only one place: causal horizons, which are null surfaces. The Clausius input of every horizon-thermodynamic derivation is therefore natively a statement about null congruences — imposed for all null directions, it is algebraically the trace-free sector, $R_{ab} k^a k^b = 8\pi G T_{ab} k^a k^b$. Now ask what each probe of gravity actually measures. A light ray's deflection integrates the transverse gradient of $\Phi + \Psi$, the sum of the temporal and spatial potentials — exactly the combination the null-sector law writes, with no reconstruction of the full field equations in between. A slow body's acceleration reads only $\nabla\Phi$: half of the null-sector content, the temporal projection alone. In the static weak field, imposing the null law isotropically forces the two potentials equal and Newton's normalization simultaneously, so the ratio of what light feels to what slow matter feels is exactly two.

Seen this way, the factor of two inverts. The puzzle is not why light bends double; it is why slow matter feels *half*. Light is the honest probe — a null trajectory samples the null-native structure whole. A particle at rest is the degenerate one, blind to the spatial face entirely; the interpolation is the textbook $(1 + v^2/c^2)$, a coupling that grows *proportionally with v^2* , from 1 to exactly 2. The 1919 measurement, on this reading, was the first observation that gravity’s degrees of freedom live on null surfaces — the natural home of entropy.

But this only works if the two faces are locked at equal strength, and here the entropic reading earns its keep rather than borrowing GR’s. At a causal horizon the vacuum’s modular Hamiltonian is the boost generator (Bisognano–Wichmann), and that one operator is read twice: its *thermality* fixes local temperature against the Tolman relation, defining clock rates — the time face, Einstein’s 1911 half. Its *variation* obeys the entanglement first law with the single calibration $S = A/4G\hbar$, writing spatial curvature — the missing half. One generator, one calibration constant: the two sectors cannot be tuned apart, and $\gamma = 1$ is structural rather than fitted. A theorem of Takesaki (1972) extends the lock to observers: any coarse-graining whose bookkeeping does not drift under its own time evolution is forced onto modular-covariant structure, so *every operationally consistent observer at a horizon inherits the same generator*. We verified the lattice image of this on free-fermion chains: coarse-graining in the modular basis preserves the first-law response to one part in 10^4 across chain lengths, fillings, and perturbation classes, while a site-basis control misses by factors of 4 to 50. The lock is basis-selective in exactly the way the theorem demands. Away from entanglement equilibrium the two faces do drift apart — by a computable, dimensionless amount, $\delta = S(\rho\|\rho_{\text{vac}})/S_{\text{BH}} \leq 2G\Delta E/c^4 R$, the excitation’s own gravitational radius over the horizon’s. The lock is an equilibrium theorem with a stated validity domain, not an article of faith.

We also built the smallest universe in which the whole story is computable. Take a quantum chain, keep only its ground state, and *define* distance by mutual information — no metric, no Hamiltonian in the ontology, only entanglement. Plant a localized defect. The emergent geometry stretches around it linearly in the defect’s strength, contracts when the defect’s sign flips, and responds not at all — to one part in 10^{14} — to a pure gauge excitation. The enclosed modular energy of concentric intervals, computed independently, reproduces the parabolic Rindler boost weight without being asked. It is a toy; it is also, so far as we know, the first place someone can *watch* geometry built from nothing but entropy curve in response to matter built from nothing but entanglement.

What would kill this? We keep a list, in writing. The framework predicts no gravitational decoherence at any interferometric precision, because the gravitating entropy is fine-grained and unitary — the coarse-grained, dissipative mechanisms of the 2011 entropic-gravity debate are already refuted by neutron interferometry, and we import that refutation as a design constraint. A confirmed gravity-correlated dephasing ends the program. Sharper and nearer: in a flowing-condensate sonic horizon, three quantities that current techniques can each measure — the Hawking phonon temperature, the entanglement entropy across the horizon, and the interior’s entropic-time rate, the last transplanted directly from the Birmingham analysis — must be set by one modular rate. A three-way lock test at the twenty-percent level appears feasible with existing methods, and it comes with

a mechanism-specific discriminator: a coherent, spectrum-preserving drive must leave the entanglement leg exactly untouched while an equal-energy incoherent injection moves it linearly. The wrong ordering falsifies the modular mechanism regardless of one's views on analogue gravity.

We are equally explicit about what is not claimed. Nowhere in the weak field does this framework compute a number GR does not; the factor of two is GR's number, and our claim is the structural account of why an entropic origin *forces* it. The cosmological constant is not predicted — it is the one thing the entropic input cannot write, the integration constant of its silence, which at least relocates the problem honestly. The selection of the observer partition is solved at horizons and open for cosmology; the continuum, Type III versions of our lattice statements remain to be written. This is a research program with one theorem-grade lock, one laboratory anchor at each end, one computable universe, and a short list of ways to die.

Einstein's 1911 calculation was not a mistake; it was the correct evaluation of the time face of gravity, missing its twin. The 1919 eclipse measured both faces at once and found them equal. If gravity is entropy read twice — coarse-grained exchange keeping the clocks, fine-grained entanglement keeping the distances, one generator keeping them honest — then that equality is not a coincidence of field equations but the signature of the ledger, and it has been sitting in plain sight, on photographic plates, for over a century.

Core references: Barontini, PRR 8, L022047 (2026) · Jacobson, PRL 75, 1260 (1995); PRL 116, 201101 (2016) · Faulkner et al., JHEP 03, 051 (2014) · Alonso-Serrano & Liška, IJGMMP 19, 2230002 (2022) · Bisognano & Wichmann (1976) · Takesaki (1972) · Casini, CQG 25, 205021 (2008) · de Nova et al., Nature 569, 688 (2019). Repository with full adjudication record and code available from the author.